1. The Terminal and Bash

The terminal is the window; Bash is the shell program running inside it.

You type	What it does
`echo $SHELL`	Shows which shell is running (here /bin/bash).
`history`	Lists the commands you typed earlier, numbered.
`↑ / Ctrl+R`	Recalls the previous command / searches your history.
`Tab`	Completes a file or command name.

```
     aadan@ubuntu: ~
aadan@ubuntu:~$ echo $SHELL
/bin/basH
aadan@ubuntu:~$
```

2. Command Syntax

Every command line has a command, optional options and arguments.

You type	What it does
`ls -l qoraal.txt`	Command ls, option -l (long listing), argument qoraal.txt.
`ls -a`	Option -a: also shows hidden names (those starting with a dot).
`ls --all`	The long form of -a. Same result.
`ls -a -l`	Options can be combined; ls -al does the same thing.

```
    aadan@ubuntu: ~
aadan@ubuntu:~$ mkdir dugsi
aadan@ubuntu:~$ cd dugsi
aadan@ubuntu:~/dugsi$ touch qoraal.txt
aadan@ubuntu:~/dugsi$ ls -l qoraal.txt
-rw-r--r-- 1 aadan aadan 0 Nov  3 19:19 qoraal.txt
aadan@ubuntu:~/dugsi$ ls -a
.  ..  qoraal.txt
aadan@ubuntu:~/dugsi$ ls --all
.  ..  qoraal.txt
aadan@ubuntu:~/dugsi$
```

3. Getting Help: man, --help, and info

You type	What it does
`man ls`	Opens the manual page for ls. Press q to quit.
`mkdir --help`	Prints a short usage summary and the list of options.
`info ls`	Opens the info documentation for ls.

```
     aadan@ubuntu: ~
aadan@ubuntu:~$ mkdir --help | head -n 6
Usage: mkdir [OPTION]... DIRECTORY...
Create the DIRECTORY(ies), if they do not already exist.
```

```
Mandatory arguments to long options are mandatory for short options too.
  -m, --mode=MODE   set file mode (as in chmod), not a=rwx - umask
  -p, --parents     no error if existing, make parent directories as needed,
aadan@ubuntu:~$
```

4. Quoting, Escaping, and Wildcards

Bash splits a line into words at spaces. Quotes keep words together, and wildcards match many names at once.

You type	What it does
`touch buug cusub`	Makes two files: buug and cusub.
`touch "buug cusub"`	Makes one file called buug cusub (the quotes keep it one word).
`echo '$HOME'`	Single quotes keep everything literal: prints $HOME.
`echo "$HOME"`	Double quotes still expand variables: prints /home/aadan.
`ls *.txt`	The * wildcard matches any characters, so every name ending in .txt is listed.

```
      aadan@ubuntu: ~
aadan@ubuntu:~$ mkdir dugsi
aadan@ubuntu:~$ cd dugsi
aadan@ubuntu:~/dugsi$ touch buug cusub
aadan@ubuntu:~/dugsi$ ls
buug  cusub
aadan@ubuntu:~/dugsi$ rm buug cusub
aadan@ubuntu:~/dugsi$ touch "buug cusub"
aadan@ubuntu:~/dugsi$ ls
'buug cusub'
aadan@ubuntu:~/dugsi$ echo '$HOME'
$HOME
aadan@ubuntu:~/dugsi$ echo "$HOME"
/home/aadan
aadan@ubuntu:~/dugsi$ touch xisaab.txt cilmi.txt qoraal.log
aadan@ubuntu:~/dugsi$ ls *.txt
cilmi.txt  xisaab.txt
aadan@ubuntu:~/dugsi$
```

4. Variables, PATH, and ~/.bashrc

A variable stores a value. PATH is the list of directories Bash searches to find a program.

You type	What it does
`MAADDO=linux`	Creates a variable (no spaces around =).
`echo $MAADDO`	Prints the value of the variable.
`echo $HOME $USER`	Prints two built-in variables: the home directory and the user name.
`export MAADDO`	Passes the variable on to programs started from this shell.
`echo $PATH`	Prints the directories searched for programs.
`ls /usr/bin/ls`	Shows that the ls program lives in /usr/bin.
`ls -a ~`	Lists home with hidden files. ~/.bashrc holds your Bash settings.

```
      aadan@ubuntu: ~
aadan@ubuntu:~$ MAADDO=linux
aadan@ubuntu:~$ echo $MAADDO
```

```
linux
aadan@ubuntu:~$ echo $HOME $USER
/home/aadan aadan
aadan@ubuntu:~$ export MAADDO
aadan@ubuntu:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap
/bin
aadan@ubuntu:~$ ls /usr/bin/ls
/usr/bin/ls
aadan@ubuntu:~$ ls -a ~
.   .bash_logout  .profile  Documents  Music         Public     Videos
..  .bashrc         Desktop   Downloads  Pictures  Templates
aadan@ubuntu:~$
```

5. Paths and Filenames

An absolute path starts with /. A relative path is read from the current directory. . is here, .. is the parent, ~ is home.

You type	What it does
`cd ~/dugsi/xisaab`	Goes to the xisaab directory (works from anywhere).
`cd ..`	Up one level to the parent directory.
`cd ../..`	Up two levels.
`cd .`	Stays where you are (. is the current directory).

```
     aadan@ubuntu: ~
aadan@ubuntu:~$ mkdir -p dugsi/xisaab
aadan@ubuntu:~$ cd dugsi/xisaab
aadan@ubuntu:~/dugsi/xisaab$ pwd
/home/aadan/dugsi/xisaab
aadan@ubuntu:~/dugsi/xisaab$ cd ..
aadan@ubuntu:~/dugsi$ cd ../..
aadan@ubuntu:/home$ cd ~/dugsi/xisaab
aadan@ubuntu:~/dugsi/xisaab$ cd .
aadan@ubuntu:~/dugsi/xisaab$ pwd
/home/aadan/dugsi/xisaab
aadan@ubuntu:~/dugsi/xisaab$
```

6. Navigating with pwd, cd, and ls

pwd shows where you are, cd moves you, ls looks around.

You type	What it does
`pwd`	Nowhere: prints the current directory.
`cd /var/log`	Absolute path: straight to /var/log.
`cd Documents`	Relative path: into Documents below here.
`cd ..`	Up one level to the parent.
`cd` or `cd ~`	Home: /home/aadan.
`cd -`	Back to the previous directory (and prints it).
`ls -a / -l / -h`	All names / long listing / human-readable sizes (ls -lh).

```
     aadan@ubuntu: ~
aadan@ubuntu:~$ pwd
/home/aadan
```

```
aadan@ubuntu:~$ cd /var/log
aadan@ubuntu:/var/log$ cd ..
aadan@ubuntu:/var$ cd -
/var/log
aadan@ubuntu:/var/log$ cd
aadan@ubuntu:~$
```

```
     aadan@ubuntu: ~
aadan@ubuntu:~$ mkdir dugsi
aadan@ubuntu:~$ cd dugsi
aadan@ubuntu:~/dugsi$ touch qoraal.txt xisaab.txt
aadan@ubuntu:~/dugsi$ mkdir buugaag
aadan@ubuntu:~/dugsi$ ls -lh
total 0
drwxr-xr-x 2 aadan aadan 40 Nov  3 19:19 buugaag
-rw-r--r-- 1 aadan aadan  0 Nov  3 19:19 qoraal.txt
-rw-r--r-- 1 aadan aadan  0 Nov  3 19:19 xisaab.txt
aadan@ubuntu:~/dugsi$ ls -a
.  ..  buugaag       qoraal.txt  xisaab.txt
aadan@ubuntu:~/dugsi$
```

7. Creating and Managing Files

You type	What it does
`touch qoraal.txt`	Creates an empty file.
`mkdir buugaag`	Creates a directory. mkdir -p a/b creates nested directories in one step.
`cp qoraal.txt qoraal.bak`	Copies a file (cp -r for directories).
`mv qoraal.bak hore.txt`	Renames (or moves) a file.
`rm -i hore.txt`	Removes a file, asking first. There is no recycle bin.
`rmdir buugaag`	Removes an empty directory only.

```
     aadan@ubuntu: ~
aadan@ubuntu:~$ mkdir dugsi
aadan@ubuntu:~$ cd dugsi
aadan@ubuntu:~/dugsi$ touch qoraal.txt
aadan@ubuntu:~/dugsi$ mkdir buugaag
aadan@ubuntu:~/dugsi$ cp qoraal.txt qoraal.bak
aadan@ubuntu:~/dugsi$ mv qoraal.bak hore.txt
aadan@ubuntu:~/dugsi$ ls
buugaag  hore.txt  qoraal.txt
aadan@ubuntu:~/dugsi$ rm -i hore.txt
rm: remove regular empty file 'hore.txt'? y
aadan@ubuntu:~/dugsi$ rmdir buugaag
aadan@ubuntu:~/dugsi$ ls
qoraal.txt
aadan@ubuntu:~/dugsi$
```

8. Lab: Navigating and Managing Files

Work done in a fresh directory called ~/ch2lab. The last column records what each step printed.

Part A: build and navigate

#	Type this	What it should do	What you saw
1	`cd`	Takes you home from wherever you are.	No output. The prompt shows ~.

#	Type this	What it should do	What you saw
2	`mkdir -p ~/ch2lab/notes`	Creates ch2lab and notes inside it in one step.	No output. Both directories created.
3	`mkdir ~/ch2lab/backup`	Adds a second directory next to notes.	No output. backup created.
4	`cd ~/ch2lab`	The prompt now ends in ~/ch2lab. Check with pwd.	Prompt shows ~/ch2lab. pwd printed /home/aadan/ch2lab.
5	`touch notes/day1.txt notes/day2.txt "notes/week 3.txt"`	Three empty files; the quotes keep "week 3" as one name.	No output. Three files created.
6	`ls -l notes`	Long listing: three lines, each starting with -.	total 0, then three lines: day1.txt, day2.txt, 'week 3.txt'.
7	`ls -a ~`	Your home again, now with hidden names. Find .bashrc.	Hidden names shown (.bash_logout, .bashrc, .profile). Found .bashrc.

```
     aadan@ubuntu: ~
aadan@ubuntu:~$ cd
aadan@ubuntu:~$ mkdir -p ~/ch2lab/notes
aadan@ubuntu:~$ mkdir ~/ch2lab/backup
aadan@ubuntu:~$ cd ~/ch2lab
aadan@ubuntu:~/ch2lab$ pwd
/home/aadan/ch2lab
aadan@ubuntu:~/ch2lab$ touch notes/day1.txt notes/day2.txt "notes/week 3.txt"
aadan@ubuntu:~/ch2lab$ ls -l notes
total 0
-rw-r--r-- 1 aadan aadan 0 Nov  3 19:19  day1.txt
-rw-r--r-- 1 aadan aadan 0 Nov  3 19:19  day2.txt
-rw-r--r-- 1 aadan aadan 0 Nov  3 19:19 'week 3.txt'
aadan@ubuntu:~/ch2lab$ ls -a ~
.   .bash_logout  .profile  Documents  Music         Public     Videos
..  .bashrc          Desktop   Downloads  Pictures  Templates  ch2lab
aadan@ubuntu:~/ch2lab$
```

Part B: copy, move, remove, look things up

#	Type this	What it should do	What you saw
8	`cp notes/day1.txt backup/`	A copy of day1.txt now sits in backup.	No output. day1.txt copied into backup.
9	`mv notes/day2.txt notes/monday.txt`	Renames day2.txt; ls notes shows monday.txt.	ls notes shows day1.txt, monday.txt, 'week 3.txt'.
10	`cp -r notes notes-copy`	Copies the whole directory with its three files.	No output. notes-copy created with three files.
11	`rm -i notes-copy/*`	Asks three times; answer y each time.	Asked three times; answered y each time.
12	`rmdir notes-copy`	Works now, because the directory is empty.	No output. notes-copy removed.
13	`echo $PATH`	Find /usr/bin in the list. Then try ls /usr/bin/mkdir	/usr/bin is in the list. ls /usr/bin/mkdir printed /usr/bin/mkdir.
14	`man rmdir`	Find the option that also removes empty parents. q to quit.	Found -p, --parents (removes the directory and its empty parents).

```
     aadan@ubuntu: ~
```

```
aadan@ubuntu:~$ mkdir -p ~/ch2lab/notes ~/ch2lab/backup
aadan@ubuntu:~$ cd ~/ch2lab
aadan@ubuntu:~/ch2lab$ touch notes/day1.txt notes/day2.txt "notes/week 3.txt"
aadan@ubuntu:~/ch2lab$ cp notes/day1.txt backup/
aadan@ubuntu:~/ch2lab$ mv notes/day2.txt notes/monday.txt
aadan@ubuntu:~/ch2lab$ ls notes
 day1.txt   monday.txt        'week 3.txt'
aadan@ubuntu:~/ch2lab$ cp -r notes notes-copy
aadan@ubuntu:~/ch2lab$ rm -i notes-copy/*
rm: remove regular empty file 'notes-copy/day1.txt'? y
rm: remove regular empty file 'notes-copy/monday.txt'? y
rm: remove regular empty file 'notes-copy/week 3.txt'? y
aadan@ubuntu:~/ch2lab$ rmdir notes-copy
aadan@ubuntu:~/ch2lab$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap
/bin
aadan@ubuntu:~/ch2lab$ ls /usr/bin/mkdir
/usr/bin/mkdir
aadan@ubuntu:~/ch2lab$ rmdir --help | grep -i parents
  -p, --parents     remove DIRECTORY and its ancestors;
aadan@ubuntu:~/ch2lab$
```

9. Practice Workspace

A temporary directory called ~/linux-ch3-lab holds the practice files.

You type	What it does
`mkdir -p project/src`	Creates a directory and its subdirectory in one step.
`printf '...'>file`	Writes text to a file (replaces it if it exists).
`printf '...'>>file`	Appends text to the end of a file.

```
    aadan@ubuntu: ~
aadan@ubuntu:~$ mkdir -p ~/linux-ch3-lab
aadan@ubuntu:~$ cd ~/linux-ch3-lab
aadan@ubuntu:~/linux-ch3-lab$ printf 'Ali\nHodan\nYusuf\n'>names.txt
aadan@ubuntu:~/linux-ch3-lab$ printf 'Mogadishu\nHargeisa\n'>cities.txt
aadan@ubuntu:~/linux-ch3-lab$ printf 'Mogadishu\nGarowe\nMogadishu\n'>>cities.txt
aadan@ubuntu:~/linux-ch3-lab$ printf 'INFO start\nERROR disk full\n'>app.log
aadan@ubuntu:~/linux-ch3-lab$ printf 'WARN retry\nERROR login failed\n'>>app.log
aadan@ubuntu:~/linux-ch3-lab$ mkdir -p project/src
aadan@ubuntu:~/linux-ch3-lab$ printf 'Linux project\n'>project/README.md
aadan@ubuntu:~/linux-ch3-lab$ printf 'Linux tools help. Linux is useful.\n'>notes.txt
aadan@ubuntu:~/linux-ch3-lab$ printf 'gray\ngrey\n'>words.txt
aadan@ubuntu:~/linux-ch3-lab$ printf '42\n100\n7\n'>scores.txt
aadan@ubuntu:~/linux-ch3-lab$                                                    printf
'ali:student:bash\nhodan:student:sh\nyusuf:admin:bash\n'>users.txt
aadan@ubuntu:~/linux-ch3-lab$ printf 'mode=practice\n'>project/config.txt
aadan@ubuntu:~/linux-ch3-lab$ printf '# TODO: test\n'>project/src/app.py
aadan@ubuntu:~/linux-ch3-lab$ ls
app.log     names.txt  project          users.txt
cities.txt  notes.txt  scores.txt  words.txt
aadan@ubuntu:~/linux-ch3-lab$
```

10. Viewing and Searching Files

You type	What it does
`cat names.txt`	Shows the contents of a file. cat -n adds line numbers.
`less long.txt  /  more`	Shows a long file page by page (less: Space, b, /word, n, q).
`head -n 2  /  tail -n 2`	Shows the first / last lines of a file (10 by default).

You type	What it does
`tail -f app.log`	Follows new lines added to a file in real time (Ctrl+C to stop).
`file / stat / ls -l`	The real type of a file / detailed information / basic information.
`which / type / command -v / whereis`	Shows where a command comes from: program, shell builtin, or alias.
`find . -name '*.log'`	Finds files by name, type (-type f/d), size (-size +10c) or time (-mmin).
`locate / updatedb`	Fast database search; updatedb refreshes the database (may need admin rights).
`grep 'Ali' names.txt`	Searches for text inside files. -n line numbers, -i ignore case, -v invert, -r recursive.
`^ $ . * [ae]`	Regex: ^ start of line, $ end of line, . any one character, * repeat, [ae] one of a or e.

```
      aadan@ubuntu: ~
aadan@ubuntu:~/linux-ch3-lab$ cat names.txt
Ali
Hodan
Yusuf
aadan@ubuntu:~/linux-ch3-lab$ cat -n project/README.md
     1        Linux project
aadan@ubuntu:~/linux-ch3-lab$ seq 1 60>long.txt
aadan@ubuntu:~/linux-ch3-lab$ less long.txt
[Space = next page, b = back, /40 = search, q = quit]
aadan@ubuntu:~/linux-ch3-lab$ head -n 2 names.txt
Ali
Hodan
aadan@ubuntu:~/linux-ch3-lab$ tail -n 2 names.txt
Hodan
Yusuf
aadan@ubuntu:~/linux-ch3-lab$ tail -n 2 -f app.log
WARN retry
ERROR login failed
[in another terminal: echo 'INFO done'>>app.log]
INFO done
[press Ctrl+C to stop]
aadan@ubuntu:~/linux-ch3-lab$
```

```
     aadan@ubuntu: ~
aadan@ubuntu:~/linux-ch3-lab$ file names.txt
names.txt: ASCII text
aadan@ubuntu:~/linux-ch3-lab$ ls -l names.txt
-rw-r--r-- 1 aadan aadan 16 Nov  6 17:08 names.txt
aadan@ubuntu:~/linux-ch3-lab$ stat names.txt
  File: names.txt
  Size: 16              Blocks: 8          IO Block: 4096   regular file
Device: 0,45    Inode: 7           Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/  aadan)   Gid: ( 1000/  aadan)
Access: 2026-11-06 17:08:10.714431533 +0000
Modify: 2026-11-06 17:08:10.711319893 +0000
Change: 2026-11-06 17:08:10.711319893 +0000
 Birth: 2026-11-06 17:08:10.709260067 +0000
aadan@ubuntu:~/linux-ch3-lab$ command -v ls
/usr/bin/ls
aadan@ubuntu:~/linux-ch3-lab$ whereis tar
tar: /usr/bin/tar
aadan@ubuntu:~/linux-ch3-lab$ type cd
cd is a shell builtin
aadan@ubuntu:~/linux-ch3-lab$ which ls
/usr/bin/ls
```

```
aadan@ubuntu:~/linux-ch3-lab$
```

```
   aadan@ubuntu: ~
aadan@ubuntu:~/linux-ch3-lab$ find . -name 'names.txt'
./names.txt
aadan@ubuntu:~/linux-ch3-lab$ find . -name '*.log'
./app.log
aadan@ubuntu:~/linux-ch3-lab$ find project -type d
project
project/src
aadan@ubuntu:~/linux-ch3-lab$ find . -name 'names.txt' -size +10c
./names.txt
aadan@ubuntu:~/linux-ch3-lab$ find project -name "*.py" -type f
project/src/app.py
aadan@ubuntu:~/linux-ch3-lab$ touch recent.txt
aadan@ubuntu:~/linux-ch3-lab$ find . -name 'recent.txt'
./recent.txt
aadan@ubuntu:~/linux-ch3-lab$
```

```
     aadan@ubuntu: ~
aadan@ubuntu:~/linux-ch3-lab$ grep 'Ali' names.txt
Ali
aadan@ubuntu:~/linux-ch3-lab$ grep -n 'ERROR' app.log
2:ERROR disk full
4:ERROR login failed
aadan@ubuntu:~/linux-ch3-lab$ grep -v 'ERROR' app.log
INFO start
WARN retry
INFO done
aadan@ubuntu:~/linux-ch3-lab$ grep -rn 'TODO' project
project/src/app.py:1:# TODO: test
aadan@ubuntu:~/linux-ch3-lab$ grep '^Ali' names.txt
Ali
aadan@ubuntu:~/linux-ch3-lab$ grep 'failed$' app.log
ERROR login failed
aadan@ubuntu:~/linux-ch3-lab$ grep 'gr[ae]y' words.txt
gray
grey
aadan@ubuntu:~/linux-ch3-lab$ grep -in 'error' app.log
2:ERROR disk full
4:ERROR login failed
aadan@ubuntu:~/linux-ch3-lab$
```

11. Redirection, Pipes, and Text Processing

STDIN (0) is input, STDOUT (1) is normal output, STDERR (2) is error messages.

You type	What it does
`>/>>`	>replaces the file;>>appends to the end.
`2>err.txt`	Saves error messages to a file (2>/dev/null discards them).
`<file`	Feeds a file to a command as input (stdin).
`|`	Pipe: sends the output of one command to the next.
`;&&||`	; runs both commands;&&only if the first succeeds; || only if it fails.
`sort -n / -r`	Sorts lines (-n numerically, -r in reverse).
`uniq -c`	Removes adjacent duplicate lines (-c counts them). Use sort first.
`cut -d: -f1`	Extracts columns (-d delimiter, -f field).

You type	What it does
`paste a b`	Joins lines of several files side by side (-d delimiter, -s serial).
`wc -l / -w / -c`	Counts lines / words / bytes.
`tr 'a-z' 'A-Z'`	Translates characters (-d delete, -s squeeze repeats). Does not take file names; use<or |.

```
aadan@ubuntu: ~
aadan@ubuntu:~/linux-ch3-lab$ ls names.txt missing.txt
ls: cannot access 'missing.txt': No such file or directory
names.txt
aadan@ubuntu:~/linux-ch3-lab$ printf 'First\n'>actions.txt
aadan@ubuntu:~/linux-ch3-lab$ printf 'Second\n'>>actions.txt
aadan@ubuntu:~/linux-ch3-lab$ cat actions.txt
First
Second
aadan@ubuntu:~/linux-ch3-lab$ ls names.txt missing.txt>out.txt 2>err.txt
aadan@ubuntu:~/linux-ch3-lab$ cat out.txt
ls: cannot access 'missing.txt': No such file or directory
names.txt
aadan@ubuntu:~/linux-ch3-lab$ cat err.txt
aadan@ubuntu:~/linux-ch3-lab$ wc -l names.txt
3 names.txt
aadan@ubuntu:~/linux-ch3-lab$ wc -l<names.txt
3
aadan@ubuntu:~/linux-ch3-lab$ tr 'a-z' 'A-Z'<names.txt
ALI
HODAN
YUSUF
aadan@ubuntu:~/linux-ch3-lab$
```

```
aadan@ubuntu: ~
aadan@ubuntu:~/linux-ch3-lab$ grep 'ERROR' app.log | wc -l
2
aadan@ubuntu:~/linux-ch3-lab$ cat names.txt | wc -l
3
aadan@ubuntu:~/linux-ch3-lab$ grep 'Mogadishu' cities.txt | wc -l
3
aadan@ubuntu:~/linux-ch3-lab$ printf 'one\n'; printf 'two\n'
one
two
aadan@ubuntu:~/linux-ch3-lab$ mkdir -p scratch&&echo OK>scratch/status.txt
aadan@ubuntu:~/linux-ch3-lab$ cat scratch/status.txt
OK
aadan@ubuntu:~/linux-ch3-lab$ test -f missing.txt || echo "missing.txt not found"
missing.txt not found
aadan@ubuntu:~/linux-ch3-lab$
```

```
aadan@ubuntu: ~
aadan@ubuntu:~/linux-ch3-lab$ sort scores.txt
100
42
7
aadan@ubuntu:~/linux-ch3-lab$ sort -n scores.txt
7
42
100
aadan@ubuntu:~/linux-ch3-lab$ sort -rn scores.txt
100
42
```

```
7
aadan@ubuntu:~/linux-ch3-lab$ sort cities.txt | uniq
Garowe
Hargeisa
Mogadishu
aadan@ubuntu:~/linux-ch3-lab$ sort cities.txt | uniq -c
      1 Garowe
      1 Hargeisa
      3 Mogadishu
aadan@ubuntu:~/linux-ch3-lab$ cut -d: -f1 users.txt
ali
hodan
yusuf
aadan@ubuntu:~/linux-ch3-lab$ cut -d: -f 1,3 users.txt
ali:bash
hodan:sh
yusuf:bash
aadan@ubuntu:~/linux-ch3-lab$
```

```
    aadan@ubuntu: ~
aadan@ubuntu:~/linux-ch3-lab$ paste names.txt scores.txt
Ali        42
Hodan        100
Yusuf        7
aadan@ubuntu:~/linux-ch3-lab$ paste -d, names.txt scores.txt
Ali,42
Hodan,100
Yusuf,7
aadan@ubuntu:~/linux-ch3-lab$ paste -s -d, names.txt
Ali,Hodan,Yusuf
aadan@ubuntu:~/linux-ch3-lab$ wc -w notes.txt
6 notes.txt
aadan@ubuntu:~/linux-ch3-lab$ echo 'too    many' | tr -s ' '
too many
aadan@ubuntu:~/linux-ch3-lab$ cut -d: -f3 users.txt | sort | uniq -c
      2 bash
      1 sh
aadan@ubuntu:~/linux-ch3-lab$ tr -cs '[:alpha:]' '\n'<notes.txt | tr '[:upper:]' '[:lower:]' |
sort | uniq -c | sort -nr | head -n 1
      2 linux
aadan@ubuntu:~/linux-ch3-lab$
```

12. Archives and Compression

An archive bundles many files into one; compression makes it smaller.

You type	What it does
`tar -cf / -cvf`	Bundles files or directories into an archive (-c create, -f name, -v verbose).
`tar -tf / -xf -C`	Lists / extracts an archive (-C target directory).
`tar -czf -cjf -cJf`	Archive and compress at once: gzip (.tar.gz), bzip2 (.tar.bz2), xz (.tar.xz).
`gzip -k / gunzip -c`	Compresses a single file (-k keeps the original; -9 best compression).
`bzip2 / xz (-k), bzcat / xzcat`	Other compressors; bzcat and xzcat read without extracting.
`zip -r / unzip -l -d`	ZIP works on Windows and macOS. -l lists, -d sets the target directory.

```
    aadan@ubuntu: ~
aadan@ubuntu:~/linux-ch3-lab$ tar -cf notes.tar notes.txt words.txt
aadan@ubuntu:~/linux-ch3-lab$ tar -cvf project.tar project/
```

```
project/
project/config.txt
project/README.md
project/src/
project/src/app.py
aadan@ubuntu:~/linux-ch3-lab$ tar -tf project.tar
project/
project/config.txt
project/README.md
project/src/
project/src/app.py
aadan@ubuntu:~/linux-ch3-lab$ mkdir -p restored-tar
aadan@ubuntu:~/linux-ch3-lab$ tar -xf project.tar -C restored-tar
aadan@ubuntu:~/linux-ch3-lab$ tar -czf project.tar.gz project/
aadan@ubuntu:~/linux-ch3-lab$ tar -cjf project.tar.bz2 project/
aadan@ubuntu:~/linux-ch3-lab$ tar -cJf project.tar.xz project/
aadan@ubuntu:~/linux-ch3-lab$ ls project.tar*
project.tar  project.tar.bz2  project.tar.gz  project.tar.xz
aadan@ubuntu:~/linux-ch3-lab$
```

```
     aadan@ubuntu: ~
aadan@ubuntu:~/linux-ch3-lab$ gzip -k -9 notes.txt
aadan@ubuntu:~/linux-ch3-lab$ ls notes.txt*
notes.txt  notes.txt.gz
aadan@ubuntu:~/linux-ch3-lab$ gunzip -c notes.txt.gz
Linux tools help. Linux is useful.
aadan@ubuntu:~/linux-ch3-lab$ bzip2 -k app.log
aadan@ubuntu:~/linux-ch3-lab$ bzcat app.log.bz2 | head -n 1
INFO start
aadan@ubuntu:~/linux-ch3-lab$ xz -k app.log
aadan@ubuntu:~/linux-ch3-lab$ xzcat app.log.xz | head -n 1
INFO start
aadan@ubuntu:~/linux-ch3-lab$ zip -r project.zip project/
  adding: project/ (stored 0%)
  adding: project/config.txt (stored 0%)
  adding: project/README.md (stored 0%)
  adding: project/src/ (stored 0%)
  adding: project/src/app.py (stored 0%)
aadan@ubuntu:~/linux-ch3-lab$ unzip -l project.zip
Archive:  project.zip
  Length      Date    Time    Name
---------  ---------- -----   ----
        0  2026-11-06 17:08   project/
       14  2026-11-06 17:08   project/config.txt
       14  2026-11-06 17:08   project/README.md
        0  2026-11-06 17:08   project/src/
       13  2026-11-06 17:08   project/src/app.py
---------                     -------
       41                     5 files
aadan@ubuntu:~/linux-ch3-lab$ unzip project.zip -d restored-zip
Archive:  project.zip
   creating: restored-zip/project/
 extracting: restored-zip/project/config.txt
 extracting: restored-zip/project/README.md
   creating: restored-zip/project/src/
 extracting: restored-zip/project/src/app.py
aadan@ubuntu:~/linux-ch3-lab$
```

13. Text Editors: Nano and Vim

Both run inside the terminal, with no graphical window.

You type	What it does

You type	What it does
`nano file`	Simple editor: Ctrl+O save, Enter confirm, Ctrl+X exit, Ctrl+W search, Ctrl+K cut line, Ctrl+U paste.
`vim file / vi / vimtutor`	Vim has three modes: Normal (default), Insert (i to type), Command (: commands).
`i Esc :w :q :wq :q!`	i insert; Esc back to Normal; :w save; :q quit; :wq save and quit; :q! quit without saving.

```
     aadan@ubuntu: ~
aadan@ubuntu:~/linux-ch3-lab$ nano project/README.md
[add a second line: Practicing Nano]
[press Ctrl+O, Enter, then Ctrl+X]
aadan@ubuntu:~/linux-ch3-lab$ cat project/README.md
Linux project
Practicing Nano
aadan@ubuntu:~/linux-ch3-lab$ vim vim-practice.txt
[press i; type: Vim practice]
[press Esc; type :wq then Enter]
aadan@ubuntu:~/linux-ch3-lab$ cat vim-practice.txt
Vim practice
aadan@ubuntu:~/linux-ch3-lab$ vim project/config.txt
[press i; make a change; press Esc; type :q then Enter]
E37: No write since last change (add ! to override)
[type :q! then Enter to quit without saving]
aadan@ubuntu:~/linux-ch3-lab$
```

14. Hard Links and Symbolic Links

A hard link is a second name for the same data; a symbolic link is a small file that holds a path.

You type	What it does
`ln a b`	Hard link: a second name for the same inode; the data is shared (same filesystem only).
`ln -s a b`	Symbolic link: stores the target path; it breaks if the target is renamed or removed.
`ls -li`	Shows the inode and link count (a hard link makes it 2).

```
     aadan@ubuntu: ~
aadan@ubuntu:~/linux-ch3-lab$ cp notes.txt link-source.txt
aadan@ubuntu:~/linux-ch3-lab$ ln link-source.txt link-hard.txt
aadan@ubuntu:~/linux-ch3-lab$ ls -li link-source.txt link-hard.txt
49 -rw-r--r-- 2 aadan aadan 35 Nov  6 17:08 link-hard.txt
49 -rw-r--r-- 2 aadan aadan 35 Nov  6 17:08 link-source.txt
aadan@ubuntu:~/linux-ch3-lab$ echo "added">>link-hard.txt
aadan@ubuntu:~/linux-ch3-lab$ tail -n 1 link-source.txt
added
aadan@ubuntu:~/linux-ch3-lab$ ln -s link-source.txt link-shortcut.txt
aadan@ubuntu:~/linux-ch3-lab$ ls -l link-shortcut.txt
lrwxrwxrwx 1 aadan aadan 15 Nov  6 17:08 link-shortcut.txt ->link-source.txt
aadan@ubuntu:~/linux-ch3-lab$ cat link-shortcut.txt
Linux tools help. Linux is useful.
added
aadan@ubuntu:~/linux-ch3-lab$ mv link-source.txt link-renamed.txt
aadan@ubuntu:~/linux-ch3-lab$ cat link-shortcut.txt
cat: link-shortcut.txt: No such file or directory
aadan@ubuntu:~/linux-ch3-lab$
```

15. Processes and Jobs

Every command you type becomes a process with a PID; the process that started it is its parent (PPID).

You type	What it does
`echo $$`	Prints the PID of your shell.
`ps  /  ps -f`	A snapshot of this terminal's processes; -f adds owner, PPID and start time.
`ps -p 1 -o pid,comm`	One PID with the columns you choose (PID 1 is systemd).
`ps aux  /  ps -ef`	Every process on the system; aux shows CPU and memory, -ef shows PPID.
`pstree -p $$`	Draws processes as a tree (parents and children).
`top  /  htop`	Live view of processes, busiest first (q to quit; htop must be installed).
`command&`	Runs a command in the background; the prompt returns at once (prints [job] and PID).
`Ctrl+C  /  Ctrl+Z`	Ctrl+C stops the job (SIGINT); Ctrl+Z pauses it (Stopped) and keeps it in the job list.
`jobs -l  /  bg %1  /  fg %2`	List this shell's jobs / resume in background / bring to foreground.

```
    aadan@ubuntu: ~
aadan@ubuntu:~$ echo $$
2318
aadan@ubuntu:~$ ps -o pid,ppid,comm
PID    PPID COMMAND
2318    2301 bash
2455    2318 ps
aadan@ubuntu:~$ ps -p 1 -o pid,comm
PID COMMAND
1 systemd
aadan@ubuntu:~$ ps
PID TTY          TIME CMD
2318 pts/0    00:00:00 bash
2460 pts/0    00:00:00 ps
aadan@ubuntu:~$ ps -f
UID          PID    PPID  C STIME TTY          TIME CMD
student     2318    2301  0 08:38 pts/0    00:00:00 bash
student     2461    2318  0 08:46 pts/0    00:00:00 ps -f
aadan@ubuntu:~$ ps -p 1 -o pid,user,comm
PID USER     COMMAND
1 root     systemd
aadan@ubuntu:~$ pstree -p $$
bash(2318)───pstree(2530)
aadan@ubuntu:~$ ps -ef | head -n 4
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 08:02 ?        00:00:02 /sbin/init
root           2       0  0 08:02 ?        00:00:00 [kthreadd]
root           3       2  0 08:02 ?        00:00:00 [rcu_gp]
aadan@ubuntu:~$
```

```
      aadan@ubuntu: ~
aadan@ubuntu:~$ ps aux | less
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.3  23380 13824 ?        Ss   08:02   0:02 /sbin/init splash
root         812  0.0  0.0   7196  2816 ?        Ss   08:02   0:00 /usr/sbin/cron -f -P
student     2318  0.0  0.1  11504  5376 pts/0    Ss   08:38   0:00 bash
[Space = next page, / = search, q = quit]
aadan@ubuntu:~$ ps aux --sort=-%mem | head -n 5
[the five biggest memory users, largest first]
aadan@ubuntu:~$ top
top - 10:43:07 up  2:41,  1 user,  load average: 0.08, 0.12, 0.10
Tasks: 231 total,   1 running, 230 sleeping,   0 stopped,   0 zombie
```

```
%Cpu(s):  1.3 us,  0.7 sy,  0.0 ni, 97.8 id,  0.0 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem :   3916.0 total,   1287.4 free,   1402.6 used,   1504.2 buff/cache
PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
1873 student   20   0 4720312 352140 128804 S   2.3   8.8   1:12.40 gnome-shell
[press q to quit top]
aadan@ubuntu:~$ top -u student
[only your own processes]
aadan@ubuntu:~$ command -v htop
/usr/bin/htop
aadan@ubuntu:~$ htop -u student
[F10 or q to quit]
aadan@ubuntu:~$
```

```
     aadan@ubuntu: ~
aadan@ubuntu:~$ sleep 30&
[1] 2541
aadan@ubuntu:~$ gnome-calculator&
[2] 2547
aadan@ubuntu:~$ sleep 100
^C
aadan@ubuntu:~$ sleep 100
^Z
[1]+  Stopped                 sleep 100
aadan@ubuntu:~$ top
^Z
[2]+  Stopped                 top
aadan@ubuntu:~$ jobs -l
[1]-  2541 Stopped                 sleep 100
[2]+  2549 Stopped                 top
aadan@ubuntu:~$ bg %1
[1]- sleep 100&
aadan@ubuntu:~$ fg %2
[top fills the screen again; press q]
aadan@ubuntu:~$ fg
sleep 100
[Ctrl+C ends it]
aadan@ubuntu:~$
```

16. Signals, kill, and Priority

A signal is a short message the kernel delivers to a process: finish up, reload, or end now.

You type	What it does
`kill PID`	Sends SIGTERM (15): a polite request to finish.
`kill -9 PID`	SIGKILL: ends the process at once; it cannot save. Use it last.
`kill -HUP / -l`	SIGHUP (1) often means reload configuration; kill -l lists all signals.
`pgrep -a / pkill / killall`	Find / stop processes by name. Run pgrep first so you do not stop too many.
`nice -n 10 cmd`	Starts a command with niceness 10 (-20 to 19; higher is kinder, default 0).
`renice -n 15 -p PID`	Changes the niceness of a running process. Making it kinder is allowed; making it greedier needs sudo.

```
    aadan@ubuntu: ~
aadan@ubuntu:~$ kill -l
1) SIGHUP       2) SIGINT       3) SIGQUIT      4) SIGILL       5) SIGTRAP   ...
aadan@ubuntu:~$ kill -l 15
TERM
aadan@ubuntu:~$ sleep 500&
```

```
[1] 2610
aadan@ubuntu:~$ kill 2610
[1]+  Terminated              sleep 500
aadan@ubuntu:~$ kill -TERM 2612
aadan@ubuntu:~$ kill %1
aadan@ubuntu:~$ kill 2650
aadan@ubuntu:~$ ps -p 2650
PID TTY          TIME CMD
2650 pts/0    00:00:03 python3
aadan@ubuntu:~$ kill -9 2650
aadan@ubuntu:~$ pgrep -a sleep
2701 sleep 400
2705 sleep 900
aadan@ubuntu:~$ pkill -u student sleep
aadan@ubuntu:~$ killall sleep
sleep: no process found
aadan@ubuntu:~$
```

```
    aadan@ubuntu: ~
aadan@ubuntu:~$ sleep 600&
[1] 3120
aadan@ubuntu:~$ jobs -l
[1]+  3120 Running           sleep 600&
aadan@ubuntu:~$ pgrep -a sleep
3120 sleep 600
aadan@ubuntu:~$ ps -p 3120 -o pid,ppid,stat,cmd
PID    PPID STAT CMD
3120    2318 S    sleep 600
aadan@ubuntu:~$ kill 3120
[1]+  Terminated       sleep 600
aadan@ubuntu:~$ jobs
aadan@ubuntu:~$ ps -p 3120
PID TTY          TIME CMD
aadan@ubuntu:~$ nice
0
aadan@ubuntu:~$ nice -n 10 sleep 300&
[1] 3301
aadan@ubuntu:~$ ps -o pid,ni,comm -p 3301
PID  NI COMMAND
3301  10 sleep
aadan@ubuntu:~$ renice -n 15 -p 3301
3301 (process ID) old priority 10, new priority 15
aadan@ubuntu:~$ renice -n 5 -p 3301
renice: failed to set priority for 3301 (process ID):
Permission denied
aadan@ubuntu:~$ sudo renice -n 5 -p 3301
3301 (process ID) old priority 15, new priority 5
aadan@ubuntu:~$
```

17. System Information

Get the facts first: kernel, uptime, memory, CPU and disks.

You type	What it does
`uname -r  -m  -n  -a`	Kernel release / machine type / hostname / everything on one line.
`hostnamectl`	Hostname, OS, kernel, and whether the machine is virtual.
`uptime  -p  -s`	Time, how long the system has run, users, and load average (1, 5, 15 minutes).
`nproc  /  lscpu`	Number of CPUs / a full processor summary.
`free -h`	RAM and swap use; read the available column, not free.
`lsblk -e 7`	Lists disks and partitions (-e 7 hides loop devices).

You type	What it does
`/proc , /sys`	Files the kernel builds in memory, always current; /proc/PID is a folder per process; /sys maps hardware.

```
      aadan@ubuntu: ~
aadan@ubuntu:~$ uname -r
6.8.0-85-generic
aadan@ubuntu:~$ uname -m
x86_64
aadan@ubuntu:~$ uname -n
ubuntu
aadan@ubuntu:~$ uname -a
Linux ubuntu 6.8.0-85-generic #85-Ubuntu SMP ... x86_64 GNU/Linux
aadan@ubuntu:~$ hostnamectl
Static hostname: ubuntu
Chassis: vm
Virtualization: oracle
Operating System: Ubuntu 24.04.3 LTS
Kernel: Linux 6.8.0-85-generic
Architecture: x86-64
Hardware Model: VirtualBox
aadan@ubuntu:~$ hostnamectl hostname
ubuntu
aadan@ubuntu:~$ uptime
10:43:07 up  2:41,  1 user,  load average: 0.08, 0.12, 0.10
aadan@ubuntu:~$ uptime -p
up 2 hours, 41 minutes
aadan@ubuntu:~$ uptime -s
2026-10-15 08:02:11
aadan@ubuntu:~$ nproc
2
aadan@ubuntu:~$
```

```
     aadan@ubuntu: ~
aadan@ubuntu:~$ free -h
total        used        free      shared  buff/cache   available
Mem:           3.8Gi       1.4Gi       1.3Gi        41Mi       1.5Gi       2.4Gi
Swap:          3.8Gi          0B       3.8Gi
aadan@ubuntu:~$ lscpu
Architecture:             x86_64
CPU(s):                   2
Vendor ID:                GenuineIntel
Model name:               Intel(R) Core(TM) i5-10210U CPU @ 1.60GHz
Hypervisor vendor:        KVM
aadan@ubuntu:~$ lsblk -e 7
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda      8:0    0   25G  0 disk
├─sda1   8:1    0    1M  0 part
└─sda2   8:2    0   25G  0 part /
sr0     11:0    1 1024M  0 rom
aadan@ubuntu:~$ lsblk -e 7 -d -o NAME,SIZE,MODEL
NAME  SIZE MODEL
sda    25G VBOX HARDDISK
sr0  1024M VBOX CD-ROM
aadan@ubuntu:~$
```

```
      aadan@ubuntu: ~
aadan@ubuntu:~$ ls /proc
1  812  1873  2318  cpuinfo  loadavg  meminfo  uptime  version ...
aadan@ubuntu:~$ ls -l /proc/meminfo
```

```
-r--r--r-- 1 root root 0 Oct 15 10:43 /proc/meminfo
aadan@ubuntu:~$ cat /proc/uptime
9671.42 18803.90
aadan@ubuntu:~$ cat /proc/loadavg
0.08 0.12 0.10 1/412 3412
aadan@ubuntu:~$ ls /proc/$$
attr  cmdline  comm  cwd  environ  exe  fd  limits  status ...
aadan@ubuntu:~$ head -n 7 /proc/$$/status
Name:   bash
Umask:  0002
State:  S (sleeping)
Tgid:   2318
Ngid:   0
Pid:    2318
PPid:   2301
aadan@ubuntu:~$ cat /proc/1/comm
systemd
aadan@ubuntu:~$ grep -m 1 "model name" /proc/cpuinfo
model name : Intel(R) Core(TM) i5-10210U CPU @ 1.60GHz
aadan@ubuntu:~$ grep -c ^processor /proc/cpuinfo
2
aadan@ubuntu:~$ head -n 3 /proc/meminfo
MemTotal:        4010000 kB
MemFree:         1318300 kB
MemAvailable:    2496100 kB
aadan@ubuntu:~$ ls /sys
block  bus  class  dev  devices  firmware  fs  kernel  module  power
aadan@ubuntu:~$ ls /sys/block
loop0  loop1  sda  sr0
aadan@ubuntu:~$ ls /sys/class/net
enp0s3  lo
aadan@ubuntu:~$ cat /sys/class/net/enp0s3/address
08:00:27:3a:5c:91
aadan@ubuntu:~$
```

18. Services with systemd

A service is a long-running background program; systemd (PID 1) starts and watches it, and systemctl controls it.

You type	What it does
`systemctl status cron`	The first question: is it running? enabled? what did it log last?
`systemctl is-active / is-enabled`	Is it running now? / Does it start at every boot?
`sudo systemctl start stop restart`	Changes what the service is doing now (restart gives a new PID).
`sudo systemctl reload / reload-or-restart`	Asks the service to re-read its configuration without stopping; if unsupported, use restart.
`sudo systemctl enable --now / disable`	enable/disable decide the next boot; --now also starts or stops it immediately.
`systemctl list-units --type=service`	Lists services; --state=running or failed filters them.

```
     aadan@ubuntu: ~
aadan@ubuntu:~$ systemctl cat cron
# /usr/lib/systemd/system/cron.service
[Unit]
Description=Regular background program processing daemon
[Service]
ExecStart=/usr/sbin/cron -f -P $EXTRA_OPTS
```

```
aadan@ubuntu:~$ systemctl status cron
● cron.service - Regular background program processing daemon
Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
Active: active (running) since Thu 2026-10-15 08:02:11 EAT; 2h 41min ago
Main PID: 812 (cron)
Oct 15 10:17:01 ubuntu CRON[3051]: (root) CMD (cd /&&run-parts --report /etc/cron.hourly)
aadan@ubuntu:~$ systemctl is-active cron
active
aadan@ubuntu:~$ systemctl is-enabled cron
enabled
aadan@ubuntu:~$
```

```
      aadan@ubuntu: ~
aadan@ubuntu:~$ sudo systemctl stop cron
[sudo] password for student:
aadan@ubuntu:~$ systemctl is-active cron
inactive
aadan@ubuntu:~$ sudo systemctl start cron
aadan@ubuntu:~$ systemctl is-active cron
active
aadan@ubuntu:~$ sudo systemctl restart cron
aadan@ubuntu:~$ systemctl show -p MainPID cron
MainPID=3388
aadan@ubuntu:~$ sudo systemctl reload cron
Failed to reload cron.service: Job type reload is
not applicable for unit cron.service.
aadan@ubuntu:~$ sudo systemctl reload-or-restart cron
aadan@ubuntu:~$ sudo systemctl disable cron
Removed "/etc/systemd/system/multi-user.target.wants/cron.service".
aadan@ubuntu:~$ systemctl is-enabled cron
disabled
aadan@ubuntu:~$ sudo systemctl enable --now cron
Created symlink /etc/systemd/system/multi-user.target.wants/cron.service → …
aadan@ubuntu:~$ systemctl list-units --type=service --state=running
UNIT                        LOAD   ACTIVE SUB     DESCRIPTION
cron.service                loaded active running Regular background program processing daemon
rsyslog.service             loaded active running System Logging Service
systemd-journald.service    loaded active running Journal Service
aadan@ubuntu:~$ systemctl list-units --type=service --state=failed
0 loaded units listed.
aadan@ubuntu:~$
```

19. Logs, Boot, and First Troubleshooting

When something fails, Linux has usually written down why: the journal, /var/log, and kernel messages.

You type	What it does
`journalctl -n 3 / -b / -b -p err`	The journal: latest lines / since this boot / errors only.
`journalctl -u cron -n 4`	The story of one service (read it after status shows a problem).
`journalctl -xe / -f`	-xe jumps to the end with explanations; -f follows new lines.
`logger "message"`	Writes a message to the journal (a test).
`ls /var/log , tail -n 3`	Classic text logs: syslog, auth.log (logins and sudo), kern.log; tail -f follows them.
`sudo dmesg -T / journalctl -k -b`	Kernel messages with readable times (Ubuntu needs sudo).
`sudo shutdown -r +5 / -c / reboot / poweroff`	Shut down or restart cleanly; -c cancels a planned shutdown.

You type	What it does
`A calm order`	Service problem: status → journalctl -u → /etc. Slow machine: uptime → free -h → top.

```
      aadan@ubuntu: ~
aadan@ubuntu:~$ journalctl -n 3
Oct 15 10:40:01 ubuntu CRON[3290]: pam_unix(cron:session): session opened
Oct 15 10:40:01 ubuntu CRON[3291]: (root) CMD (command -v debian-sa1 ...)
Oct 15 10:40:01 ubuntu CRON[3290]: pam_unix(cron:session): session closed
aadan@ubuntu:~$ journalctl -b -p err
[only errors from this boot; q to quit]
aadan@ubuntu:~$ journalctl -u cron -n 4
Oct 15 10:31:12 ubuntu systemd[1]: Stopping cron.service - Regular b…
Oct 15 10:31:12 ubuntu systemd[1]: cron.service: Deactivated successfully.
Oct 15 10:31:12 ubuntu systemd[1]: Stopped cron.service - Regular b…
Oct 15 10:31:20 ubuntu systemd[1]: Started cron.service - Regular b…
aadan@ubuntu:~$ logger "week 8 test from student"
[appears at once in a second terminal running journalctl -f]
aadan@ubuntu:~$ ls /var/log
apt  auth.log  boot.log  dpkg.log  journal  kern.log  syslog ...
aadan@ubuntu:~$ grep sudo /var/log/auth.log | tail -n 1
2026-10-15T10:31:12.402118+03:00 ubuntu sudo:   student : TTY=pts/0 ; PWD=/home/student ;
USER=root ; COMMAND=/usr/bin/systemctl stop cron
aadan@ubuntu:~$ journalctl _COMM=sudo -n 1
Oct 15 10:31:12 ubuntu sudo[3360]:   student : TTY=pts/0 ; PWD=/home/student ; USER=root ;
COMMAND=/usr/bin/systemctl stop cron
aadan@ubuntu:~$ last -n 2
student  tty2         tty2             Thu Oct 15 08:03   still logged in
reboot   system boot  6.8.0-85-generic Thu Oct 15 08:02   still running
aadan@ubuntu:~$ dmesg
dmesg: read kernel buffer failed: Operation not permitted
aadan@ubuntu:~$ sudo dmesg -T | tail -n 2
[Thu Oct 15 08:02:14 2026] sd 2:0:0:0: [sda] Attached SCSI disk
[Thu Oct 15 08:02:19 2026] e1000 0000:00:03.0 enp0s3: NIC Link is Up
aadan@ubuntu:~$ sudo shutdown -r +5 "Rebooting for the lab"
Reboot scheduled for Thu 2026-10-15 11:05:00 EAT,
use 'shutdown -c' to cancel.
aadan@ubuntu:~$ sudo shutdown -c
[cancelled; nothing happens]
aadan@ubuntu:~$
```